

Section 7.5 - Strategy for Integration

Table of Integration Formulas Constants of integration have been omitted.

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1)$$

$$2. \int \frac{1}{x} dx = \ln |x|$$

$$3. \int e^x dx = e^x$$

$$4. \int b^x dx = \frac{b^x}{\ln b}$$

$$5. \int \sin x dx = -\cos x$$

$$6. \int \cos x dx = \sin x$$

$$7. \int \sec^2 x dx = \tan x$$

$$8. \int \csc^2 x dx = -\cot x$$

$$9. \int \sec x \tan x dx = \sec x$$

$$10. \int \csc x \cot x dx = -\csc x$$

$$11. \int \sec x dx = \ln |\sec x + \tan x|$$

$$12. \int \csc x dx = \ln |\csc x - \cot x|$$

$$13. \int \tan x dx = \ln |\sec x|$$

$$14. \int \cot x dx = \ln |\sin x|$$

$$15. \int \sinh x dx = \cosh x$$

$$16. \int \cosh x dx = \sinh x$$

$$17. \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$$

$$18. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right), \quad a > 0$$

$$*19. \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right|$$

$$*20. \int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}|$$

Example #1

Find

$$\int (\sec(x) - \cos(x))^2 dx.$$

Expand and simplify:

$$(\sec(x) - \cos(x))^2$$

$$= \sec^2(x) - 2\cos(x)\sec(x) + \cos^2(x)$$

$$= \sec^2(x) - 2 + \frac{1}{2} + \frac{1}{2}\cos(2x)$$

$$\Rightarrow \int (\sec(x) - \cos(x))^2 dx$$

$$= \int \sec^2(x) - \frac{3}{2} + \frac{1}{2}\cos(2x) dx$$

$$= \tan(x) - \frac{3}{2}x + \frac{1}{4}\sin(2x) + C$$

Example #2

Find

$$\int \frac{x^2}{\sqrt{4-x^6}} dx.$$

Let

$$u = x^3 \quad du = 3x^2 dx$$

then

$$\begin{aligned} \int \frac{x^2}{(4-x^6)^{1/2}} dx &= \frac{1}{3} \int \frac{1}{(4-u^2)^{1/2}} du \\ &= \frac{1}{6} \int \frac{1}{(1-(u/2)^2)^{1/2}} du. \end{aligned}$$

Use another substitution

$$w = \frac{u}{2} \quad dw = \frac{1}{2} du \Rightarrow du = 2dw$$

so

$$\begin{aligned} \frac{1}{6} \int \frac{1}{(1-(u/2)^2)^{1/2}} du &= \frac{1}{3} \int \frac{1}{(1-w^2)^{1/2}} dw \\ &= \frac{1}{3} \arcsin(w) + C \\ &= \frac{1}{3} \arcsin\left(\frac{u}{2}\right) + C. \end{aligned}$$

We have

$$\int \frac{x^2}{(4-x^6)^{1/2}} dx = \frac{1}{3} \arcsin\left(\frac{x^3}{2}\right) + C.$$

Example #3

Find

$$\int \frac{4 - 3x^2}{x(x^2 - 4)} dx.$$

Do we need a PFD? Expand

$$\frac{4 - 3x^2}{x(x^2 - 4)} = \frac{4 - 3x^2}{x^3 - 4x}$$

and let

$$u = x^3 - 4x \quad du = (3x^2 - 4)dx.$$

Now

$$\begin{aligned} \int \frac{4 - 3x^2}{x(x^2 - 4)} dx &= - \int \frac{1}{u} du \\ &= -\ln|u| + C \\ &= \ln\left|\frac{1}{u}\right| + C \\ &= \ln\left|\frac{1}{x^3 - 4x}\right| + C \end{aligned}$$

Example #4

Find

$$\int x e^{\sqrt{1+x^2}} dx.$$

Let

$$u = 1 + x^2 \quad du = 2x \, dx?$$

then

$$\int x e^{\sqrt{1+x^2}} \, dx = \frac{1}{2} \int e^{u^{1/2}} \, du.$$

This requires another clever substitution and integration by parts to complete...

Instead we have a $1 + x^2$ term, so consider

$$x = \tan(t) \quad dx = \sec^2(t) \, dt$$

with $t \in (-\pi/2, \pi/2)$ to ensure $\sec(t) > 0$. We get

$$\begin{aligned} \int x e^{\sqrt{1+x^2}} \, dx &= \int \tan(t) e^{(1+\tan^2(t))^{1/2}} \sec^2(t) \, dt \\ &= \int \tan(t) e^{\sec(t)} \sec^2(t) \, dt \end{aligned}$$

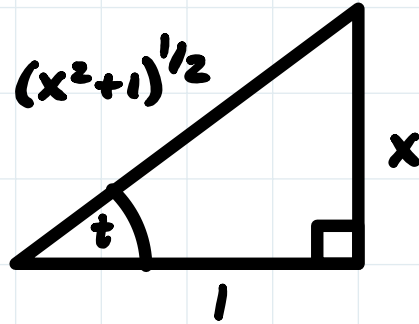
Now let

$$u = \sec(t) \quad du = \sec(t) \tan(t) \, dt$$

u 's		dv 's
u	+	e^u
1	-	e^u
0		e^u

and apply Integration by Parts

$$\begin{aligned}\int ue^u du &= (u-1)e^u + C \\ &= (\sec(t)-1)e^{\sec(t)} + C\end{aligned}$$



$$\tan(t) = x \Rightarrow \sec(t) = \frac{1}{\cos(t)} = (1+x^2)^{1/2}$$

Finally,

$$\int xe^{(x^2+1)^{1/2}} dx = ((x^2+1)^{1/2}-1)e^{(x^2+1)^{1/2}} + C.$$

Example #5

Find

$$\int \ln(1 + x^2) \, dx.$$

Integrate by parts

$$u = \ln(1+x^2) \quad dv = dx$$

$$du = \frac{2x}{1+x^2} dx \quad v = x$$

$$\int \ln(1+x^2) dx = uv - \int v du$$

$$= x \ln(1+x^2) - \int \frac{2x^2}{1+x^2} dx$$

Then decompose the rational function with polynomial division

$$\begin{array}{r} 2 \\ 1+x^2 \overline{) 2x^2} \\ \ominus \underline{2x^2+2} \\ -2 \end{array}$$

tells us

$$2(1+x^2) - 2 = 2x^2$$

and so

$$2 - \frac{2}{1+x^2} = \frac{2x^2}{1+x^2}.$$

Thus

$$\begin{aligned} \int \frac{2x^2}{1+x^2} dx &= \int 2 - \frac{2}{1+x^2} dx \\ &= 2x - 2 \arctan(x) + C \end{aligned}$$

and

$$\int \ln(1+x^2)dx = x\ln(1+x^2) - 2x + 2\arctan(x) + C.$$

Example #6

Find

$$\int x e^{\sqrt{x}} \, dx.$$

Integrate by parts?

$$u = e^{x^{1/2}} \quad dv = x dx$$

$$du = \frac{e^{x^{1/2}}}{2x^{1/2}} dx \quad v = \frac{1}{2}x^2$$

(Note: $\int e^{x^{1/2}} dx$ requires integration by parts...)

$$\int x e^{x^{1/2}} dx = \frac{1}{2}x^2 e^{x^{1/2}} - \underbrace{\int \frac{1}{2}x^2 \cdot \frac{e^{x^{1/2}}}{2x^{1/2}} dx}_{\text{more complicated}}$$

Instead,

$$u = x^{1/2}, \quad du = \frac{1}{2x^{1/2}} dx \Rightarrow 2u du = dx$$

so

$$\int x e^{x^{1/2}} dx = \int u^2 e^u \cdot 2u du = 2 \int u^3 e^u du$$

Or... rewrite

$$x e^{x^{1/2}} = x e^{x^{1/2}} \cdot \frac{2x^{1/2}}{2x^{1/2}}$$

and let

$$u = x^{1/2} \Rightarrow du = \frac{1}{2x^{1/2}} dx$$

to get

$$\int x e^{x^{1/2}} dx = 2 \int x^{3/2} e^{x^{1/2}} \cdot \frac{1}{2x^{1/2}} dx$$

$$= 2 \int u^3 e^u du$$

Now use tabular integration by parts

u's		dv's
u^3	+	e^u
$3u^2$	-	e^u
$6u$	+	e^u
6	-	e^u
0		e^u

$$2 \int u^3 e^u du = 2(u^3 - 3u^2 + 6u - 6)e^u + C$$

$$= 2(x^{3/2} - 3x + 6x^{1/2} - 6)e^{x^{1/2}} + C$$

Example #7

Find

$$\int x^3 \ln(x) \, dx.$$

Polynomial and a logarithm, integration by parts!

$$u = \ln(x), \quad dv = x^3 dx$$

$$du = \frac{1}{x} dx \quad v = \frac{1}{4}x^4$$

Thus

$$\begin{aligned} \int x^3 \ln(x) dx &= \frac{1}{4}x^4 \ln(x) - \frac{1}{4} \int x^3 dx \\ &= \frac{1}{4}x^4 \ln(x) - \frac{1}{16}x^4 + C. \end{aligned}$$

Example #8

Find

$$\int \frac{1}{x^3 + 1} dx$$

Observe

$$x^3 + 1 = 0 \text{ when } x = -1 \\ \Rightarrow x + 1 \text{ divides } x^3 + 1$$

and

$$x^3 + 1 = (x + 1)(x^2 - x + 1).$$

The form for the PFD:

$$\frac{1}{x^3 + 1} = \frac{1}{(x + 1)(x^2 - x + 1)} \\ = \frac{A}{x + 1} + \frac{Bx + C}{x^2 - x + 1}$$

so

$$1 = A(x^2 - x + 1) + (Bx + C)(x + 1) \\ = A(x^2 - x + 1) \\ + B(x^2 + x) \\ + C(x + 1).$$

Let $x = -1$

$$1 = A(1 + 1 + 1) = 3A \Rightarrow A = \frac{1}{3}$$

2nd order terms tell us:

$$0 = (A + B)x^2 \Rightarrow B = -A = -\frac{1}{3}$$

Constant terms assert:

$$1 = A + C \Rightarrow C = \frac{2}{3}.$$

We have

$$\frac{1}{x^3 + 1} = \frac{1}{3} \cdot \frac{1}{x + 1} + \frac{1}{3} \cdot \frac{-x + 2}{x^2 - x + 1}$$

$$\Rightarrow \int \frac{1}{x^3 + 1} dx = \frac{1}{3} \int \frac{1}{x + 1} dx + \frac{1}{3} \int \frac{-x + 2}{x^2 - x + 1} dx$$

To show

$$\begin{aligned} \frac{1}{3} \int \frac{-x + 2}{x^2 - x + 1} dx &= -\frac{1}{6} \ln |x^2 - x + 1| \\ &\quad + \frac{\sqrt{3}}{3} \arctan\left(\frac{2x - 1}{\sqrt{3}}\right) + C \end{aligned}$$

use the same method we used on

$$\int \frac{-x + 9}{x^2 + 2x + 5} dx$$

in § 7.4.

Hence

$$\begin{aligned} \int \frac{1}{x^3 + 1} dx &= \frac{1}{3} \ln |x + 1| - \frac{1}{6} \ln |x^2 - x + 1| \\ &\quad + \frac{\sqrt{3}}{3} \arctan\left(\frac{2x - 1}{\sqrt{3}}\right) + C. \end{aligned}$$